

CHAPTER – 4(ZEPTO E-COMMERCE MANAGEMENT SYSTEM)

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION:

The **Zepto E-Commerce Management System** is designed to efficiently manage customers, products, categories, orders, payments, deliveries, carts, and administrators. Functional dependencies define the relationships between attributes within each table, while redundancy analysis and data anomaly detection help maintain data consistency and minimize duplication. A well-designed database improves system performance, ensures data integrity, and supports efficient management of online grocery shopping operations.

1. CUSTOMER TABLE:

Functional Dependency

Customer_ID → **Customer_Name, Email, Phone_Number, Address, Password**

Redundancy Analysis

Customer information is stored only once in the Customer table. Orders and Cart tables reference customers using **Customer_ID**, preventing duplicate customer records.

Data Anomalies

No data anomalies were identified.

Reason: Customer information is uniquely identified by **Customer_ID** and linked to the **Order** and **Cart** tables using foreign keys. The normalized design prevents insert, update, and delete anomalies.

2. CATEGORY TABLE:

Functional Dependency

Category_ID → **Category_Name, Description**

Redundancy Analysis

Each category is stored only once. Products reference the category using **Category_ID** instead of repeating category information.

Data Anomalies

No data anomalies were identified.

Reason: Category information is uniquely identified by **Category_ID** and linked to the **Product** table using a foreign key. The normalized design prevents insert, update, and delete anomalies.

3. PRODUCT TABLE:

Functional Dependency

Product_ID → **Product_Name, Price, Stock, Category_ID, Description**

Redundancy Analysis

Product information is stored only in the Product table. Order Item and Cart tables reference products using **Product_ID**, avoiding duplicate product information.

Data Anomalies

No data anomalies were identified.

Reason: Product information is uniquely identified by **Product_ID** and linked to the **Cart** and **Order Item** tables using foreign keys. The normalized design prevents insert, update, and delete anomalies.

4. CART TABLE:

Functional Dependency

Cart_ID → **Customer_ID, Product_ID, Quantity, Added_Date, Total_Price**

Redundancy Analysis

The Cart table stores only cart-related information. Customer and product details are referenced using foreign keys instead of being duplicated.

Data Anomalies

No data anomalies were identified.

Reason: Cart information is uniquely identified by **Cart_ID** and linked to the **Customer** and **Product** tables using foreign keys. The normalized design prevents insert, update, and delete anomalies.

5. ORDER TABLE:

Functional Dependency

Order_ID → **Customer_ID, Order_Date, Total_Amount, Order_Status**

Redundancy Analysis

The Order table stores only order-related information. Customer information is referenced through **Customer_ID**, reducing redundancy.

Data Anomalies

No data anomalies were identified.

Reason: Order information is uniquely identified by **Order_ID** and linked to the **Customer**, **Payment**, **Delivery**, and **Order Item** tables using foreign keys. The normalized design prevents insert, update, and delete anomalies.

6. ORDER ITEM TABLE:

Functional Dependency

Order_Item_ID → **Order_ID**, **Product_ID**, **Quantity**, **Price**

Redundancy Analysis

The Order Item table connects orders and products without repeating product details in the Order table.

Data Anomalies

No data anomalies were identified.

Reason: Order item information is uniquely identified by **Order_Item_ID** and linked to the **Order** and **Product** tables using foreign keys. The normalized design prevents insert, update, and delete anomalies.

7. PAYMENT TABLE:

Functional Dependency

Payment_ID → **Order_ID**, **Payment_Method**, **Payment_Status**, **Amount**

Redundancy Analysis

Payment information is maintained separately from order details to avoid duplicate payment records.

Data Anomalies

No data anomalies were identified.

Reason: Payment information is uniquely identified by **Payment_ID** and linked to the **Order** table using a foreign key. The normalized design prevents insert, update, and delete anomalies.

8. DELIVERY TABLE:

Functional Dependency

Delivery_ID → **Order_ID**, **Delivery_Partner**, **Delivery_Status**, **Delivery_Date**

Redundancy Analysis

Delivery details are stored separately and linked through **Order_ID**, preventing repeated delivery information.

Data Anomalies

No data anomalies were identified.

Reason: Delivery information is uniquely identified by **Delivery_ID** and linked to the **Order** table using a foreign key. The normalized design prevents insert, update, and delete anomalies.

9. ADMIN TABLE:

Functional Dependency

Admin_ID → **Admin_Name**, **Email**, **Phone_Number**, **Password**

Redundancy Analysis

Administrator information is stored only once in the Admin table, preventing duplicate login information.

Data Anomalies

No data anomalies were identified.

Reason: Administrator information is uniquely identified by **Admin_ID** and stored independently. The normalized design prevents insert, update, and delete anomalies.

CONCLUSION:

The **Zepto E-Commerce Management System** is designed using proper functional dependencies and normalized tables. Each entity stores only its relevant data, minimizing redundancy and preventing insertion, update, and deletion anomalies. This database design improves data consistency, maintains integrity, and ensures efficient management of the online grocery shopping system.